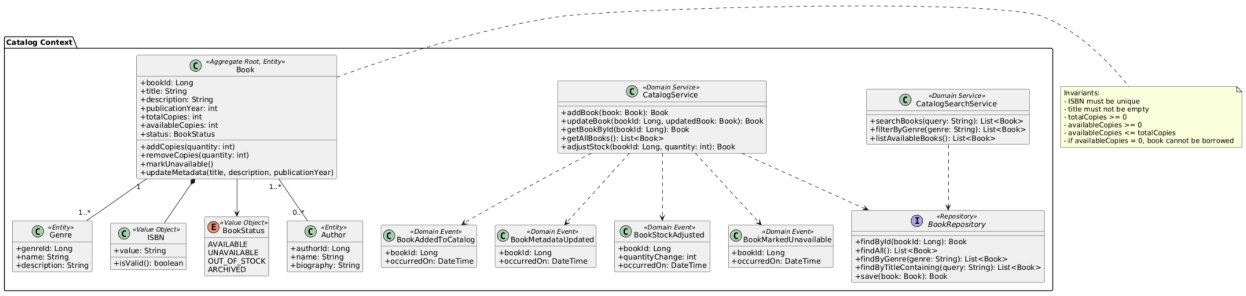

Catalog Context

Ownership and Responsibility

This context is responsible for managing the library's bibliographic information and stock visibility. Its primary purpose is to allow users to browse, search, and filter books while allowing librarians or administrators to maintain the catalog records.

The Catalog context includes responsibility for:

- storing and maintaining book metadata
- classifying books by genre and author
- exposing stock and availability information
- supporting browsing and searching of the library collection
- updating book records and stock levels

The Catalog context does not manage the borrowing lifecycle, overdue logic, fine calculation, authentication, or notifications. These concerns belong to the Loan, User/Auth, and Notification bounded contexts.

Cross-Context Dependency Notes

The Catalog context interacts with other bounded contexts in the following ways:

Bounded Context	Relationship to Catalog	Notes

Loan Context	Depends on Catalog	Loan checks stock availability before creating a loan and updates stock after borrow/return
User/Auth Context	Supports controlled access	Administrative catalog updates should only be available to authenticated staff users
Notification Context	Indirect relationship	Catalog does not directly notify users; notifications are triggered by Loan events

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Domain Modelling and Ubiquitous Language

The Catalog context uses a ubiquitous language centred around terms such as Book, Author, Genre, ISBN, Stock, Availability, Catalog Entry, and Search. These terms were consistently applied across the UML model, event storming outcomes, and implementation planning.

The context supports the following key capabilities:

- browse all books in the catalog
- search books by title or keyword
- filter books by genre
- view bibliographic details of a selected book
- update catalog metadata
- maintain stock and availability values

DDD Building Blocks

Entities

The main entities identified in the Catalog context are:

- Book – represents a catalogued library item and acts as the main aggregate root
- Author – represents the author associated with one or more books
- Genre – represents the classification used to group books

Value Objects

The following value objects were identified:

- ISBN – represents the unique bibliographic identifier for a book
- BookStatus – represents the current availability state of a book, such as available or out of stock

Aggregate

The main aggregate in this context is the Book aggregate, with Book acting as the aggregate root. The aggregate encapsulates bibliographic metadata and stock-related consistency rules.

Repository

- BookRepository – provides persistence and retrieval operations for Book aggregates

Domain Services

- CatalogService – coordinates creation, update, retrieval, and stock adjustment operations
- CatalogSearchService – handles searching and filtering operations for the catalog

Domain Events

The Catalog context includes the following domain events:

- BookAddedToCatalog
- BookMetadataUpdated
- BookStockAdjusted
- BookMarkedUnavailable

These events support traceability and provide a basis for future event-driven integration if the system evolves further.

Aggregate Design and Invariants

The Book aggregate was chosen as the main aggregate because it groups together the data and rules that must remain consistent for each catalog entry. The aggregate root controls updates to metadata and stock information, ensuring that invalid book states cannot be introduced.

Aggregate Boundary Justification

The Book aggregate includes:

- bibliographic information such as title, description, publication year, and ISBN
- classification links such as author and genre
- stock information such as total copies and available copies
- current availability state

This boundary was selected because these elements change together and must remain consistent as a single unit of business meaning. For example, when stock is adjusted, both the availability state and the available copy count may need to change together.

Invariants

The following invariants apply within the Catalog context:

- each book must have a unique ISBN
- each book must have a valid title
- `totalCopies` must never be negative
- `availableCopies` must never be negative
- `availableCopies` must never exceed `totalCopies`
- a book cannot be considered borrowable when `availableCopies = 0`

Transactional Consistency Rules

The main consistency rules for this context are:

- when a book is added, it must be persisted with valid metadata and stock values
- when stock is adjusted, the book's availability state must be updated consistently
- when a loan is created in the Loan context, the corresponding available stock in Catalog must be decremented
- when a loan is returned, the corresponding available stock in Catalog must be incremented

These rules ensure that Catalog remains the source of truth for stock and bibliographic metadata.

Inter-Context Contracts

The Catalog context publishes data and operations that are consumed by the wider system. These contracts are important because other contexts should not directly manipulate Catalog internals.

Catalog to Loan Context

The Loan context depends on Catalog to:

- retrieve book details for display
- check stock availability before allowing a borrow action
- decrement available stock when a loan is created
- increment available stock when a loan is returned

A published API for this interaction may include operations such as:

- `GET /api/books`
- `GET /api/books/{id}`
- `GET /api/books/search?query=...`
- `GET /api/books?genre=...`
- `PUT /api/books/{id}/stock`

Catalog remains the source of truth for bibliographic data and stock counts, while Loan remains the source of truth for loan status and loan period management.

Catalog to User/Auth Context

The Catalog context has a limited dependency on the User/Auth context. Browsing and searching may be available more broadly, but administrative actions such as adding or updating books should only be available to authenticated and authorised staff users. This protects catalog integrity without placing identity logic inside the Catalog context itself.

Catalog to Notification Context

The Catalog context does not directly send notifications. Notifications related to borrowing, returning, or overdue loans are initiated by the Loan context. However, the Catalog context may emit stock-related domain events that could later be consumed by other services if the architecture is expanded.

Anti-Corruption Strategy

An important anti-corruption principle is that the Loan context should not directly modify Catalog database entities. Instead, it should interact with Catalog through published APIs or explicit message contracts. This preserves the independence of each bounded context and reduces coupling between services.